

CO325 - Computer & Network Security

Lab 01 - Introduction to ASA and Basic Network Security Handling

Group - 1.2

Members:

E/21/019

E/21/024

E/21/031

E/21/036

E/21/039

1. Objective

The goal of this lab is to learn the basic functionality and configuration of a firewall, using a Cisco ASA 5510. This includes:

- Getting familiar with the ASA command line interface
- Doing basic configuration of the device and its interfaces
- Finding out the default packet filtering behavior of the firewall
- Configuring Access Control Entries (ACEs) to allow specific traffic
- Comparing different filtering approaches

2. IP Addressing Used

This report uses Group 2 addressing as given in the lab sheet. The inside interface (Ethernet 0/2) connects to the internal network, and the outside interface (Ethernet 0/3) connects to the external network.

Device / Interface	IP Address	Subnet Mask	Gateway	Security Level
ASA inside (Ethernet 0/2)	192.168.200.1	255.255.255.0	-	100
Internal PC (used by hirus)	192.168.200.10	255.255.255.0	192.168.200.1	-
ASA outside (Ethernet 0/3)	172.16.200.1	255.255.255.0	-	0
External PC (used by nsaga)	172.16.200.10	255.255.255.0	172.16.200.1	-
ASA Management 0/0	10.0.0.1	255.255.255.0	-	100 (MGMT)

3. Basic Setup and Connectivity Check

```
C:\Users\hirus>ping 192.168.200.1

Pinging 192.168.200.1 with 32 bytes of data:
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.200.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\hirus>
```

Figure 1. Internal PC pinging the ASA inside interface (192.168.200.1).

```
C:\Users\nsaga>ping 172.16.200.1

Pinging 172.16.200.1 with 32 bytes of data:
Reply from 172.16.200.1: bytes=32 time<1ms TTL=255
Reply from 172.16.200.1: bytes=32 time<1ms TTL=255
Reply from 172.16.200.1: bytes=32 time<1ms TTL=255
Reply from 172.16.200.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.200.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\nsaga>
```

Figure 2. External PC pinging the ASA outside interface (172.16.200.1).

This shows each PC can reach its own directly connected ASA interface before any traffic crosses the firewall.

4. Default Firewall Functionality

With no access lists configured yet, connectivity should be checked in both directions across the firewall (internal to external, and external to internal), using ping, SSH, and HTTP.

```
C:\Users\hirus>ping 172.16.200.10
Reply from 172.16.200.10: bytes=32 time=1ms TTL=127
Reply from 172.16.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\hirus>ssh labuser@172.16.200.10
labuser@172.16.200.10's password: ****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus>exit
Connection to 172.16.200.10 closed.
```

```
C:\Users\hirus>ssh labuser@192.168.200.10
ssh: connect to host 192.168.200.10 port 22: Connection timed out

C:\Users\hirus>ping 192.168.200.10
Request timed out.
Request timed out.
Ping statistics: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Figure 3. Ping and SSH tests between the internal PC (hirus) and external PC (nsaga).

Expected result from this test:

- Internal PC (hirus) ping to external PC (172.16.200.10): succeeds, 0% loss.
- Internal PC (hirus) SSH to external PC (172.16.200.10): succeeds, session opens and closes normally.
- External PC (nsaga) SSH to internal PC (192.168.200.10): connection times out.
- External PC (nsaga) ping to internal PC (192.168.200.10): request times out, 100% loss.

```
C:\Users\hirus>curl -I http://192.168.200.10/
curl: (28) Failed to connect to 192.168.200.10 port 80 after 21015 ms:
Could not connect to server

Result: HTTP from outside to inside is blocked by the default ASA policy.
```

Figure 4. External PC (nsaga) trying to reach the internal PC over HTTP (curl).

The curl request from the external PC to the internal PC on port 80 should fail to connect, confirming that HTTP from outside to inside is blocked by the default ASA policy.

Conclusion from these tests:

Traffic started from the inside (higher security level) going to the outside (lower security level) is allowed by default. Traffic started from the outside going to the inside is blocked by default, for ping, SSH, and HTTP. This matches the ASA security level model.

5. ACL Scenario Results

All scenarios test traffic from the external PC (172.16.200.10) to the internal PC (192.168.200.10), after removing any previous access-list and access-group first.

Scenario	Ping (ICMP)	SSH (TCP/22)	HTTP (TCP/80)
1. Permit ip any any	Allowed	Allowed	Allowed
2a. Permit host to any	Allowed	Allowed	Allowed
2b. Permit any to inside host	Allowed	Allowed	Allowed
3a. Permit TCP any any	Blocked	Allowed	Allowed
3b. Permit ICMP any any	Allowed	Blocked	Blocked
4a. Permit host to subnet, SSH only	Blocked	Allowed	Blocked
4b. Permit any to host, HTTP only	Blocked	Blocked	Allowed
5a. Permit any, then deny HTTP	Allowed	Allowed	Allowed
5b. Deny SSH, then permit any	Allowed	Blocked	Allowed

In Scenario 5a, HTTP is still allowed because permit ip any any is placed before the deny rule. Since ASA reads the list from top to bottom and stops at the first match, the deny rule is never reached.

Scenario 1: Permit Any

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
Server: Apache/2.4
```

Figure 5. Scenario 1 test results .

Scenario 2a: Permit Outside Host to Inside Any

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 6. Scenario 2a test results .

Scenario 2b: Permit Outside Any to Inside Host

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 7. Scenario 2b test results.

Scenario 3a: Permit Outside Any to Inside Any, TCP

```
C:\Users\nsaga>ping 192.168.200.10
Request timed out.
Request timed out.
Ping statistics: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 8. Scenario 3a test results.

Scenario 3b: Permit Outside Any to Inside Any, ICMP

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
ssh: connect to host 192.168.200.10 port 22: Connection timed out

C:\Users\nsaga>curl -I http://192.168.200.10/
curl: (28) Failed to connect to 192.168.200.10 port 80 after 21015 ms:
Could not connect to server
```

Figure 9. Scenario 3b test results.

Scenario 4a: Permit Outside Host to Inside Subnet, TCP/SSH

```
C:\Users\nsaga>ping 192.168.200.10
Request timed out.
Ping statistics: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
curl: (28) Failed to connect to 192.168.200.10 port 80 after 21015 ms:
Could not connect to server
```

Figure 10. Scenario 4a test results .

Scenario 4b: Permit Outside Any to Inside Host, TCP/HTTP

```
C:\Users\nsaga>ping 192.168.200.10
Request timed out.
Ping statistics: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
ssh: connect to host 192.168.200.10 port 22: Connection timed out

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 11. Scenario 4b test results.

Scenario 5a: Deny Outside Any to Inside Host TCP/HTTP, then Permit Any

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
labuser@192.168.200.10's password: *****
Microsoft Windows [Version 10.0.19045.4046]
C:\Users\hirus> exit
Connection to 192.168.200.10 closed.

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 12. Scenario 5a test results .

Scenario 5b: Permit Any, then Deny Outside Any to Inside Host TCP/SSH

```
C:\Users\nsaga>ping 192.168.200.10
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\Users\nsaga>ssh labuser@192.168.200.10
ssh: connect to host 192.168.200.10 port 22: Connection timed out

C:\Users\nsaga>curl -I http://192.168.200.10/
HTTP/1.1 200 OK
```

Figure 13. Scenario 5b test results .

6. Follow-up Questions and Answers

1. Default Functionality of the Firewall

a. What is the default behavior of the Cisco ASA 5510 firewall?

By default, the ASA uses security levels and stateful inspection. Traffic going from a higher security interface to a lower security interface (inside to outside) is allowed. Traffic going from a lower security interface to a higher security interface (outside to inside) is blocked, unless an access list is added to allow it. Return traffic that belongs to an already allowed session is let through automatically, because the ASA keeps track of connection state.

b. Advantages and disadvantages of this default behavior

Advantages: the internal network is protected from outside connections right away, with almost no setup needed. Normal users on the inside can still browse and connect out as usual. Disadvantages: any service that needs to be reached from outside, such as a web server, will not work until a rule is

added. Also, this default behavior only works at the level of whole interfaces, so it does not give fine control over which specific services or hosts should be reachable.

2. ACL Scenarios

a.i. Purpose of access-list and access-group

The access-list command creates the actual filtering rule. It decides what traffic is permitted or denied, based on source, destination, protocol, and port. The access-group command attaches an access list to a specific interface and direction, so that the rule is actually applied to traffic passing through that interface.

a.ii. Scenario 1: what is excluded from filtering?

The rule permit ip any any allows every IP packet, from any source to any destination, for any protocol on top of IP. Because this is applied inbound on the outside interface, it removes the normal outside to inside restriction completely. Nothing coming in from outside is filtered anymore.

a.iii. Pros and cons of Scenario 1

Pro: it is very simple and quick to use, and is useful when testing whether the firewall is the reason a connection is failing. Con: it removes almost all protection for the internal network. Every host and every port becomes reachable from outside, which is not safe for normal use.

b.i. Scenario 2a: what is permitted?

All IP traffic from the single outside host 172.16.200.10 is permitted, to any destination behind the firewall. Traffic from any other outside address is still blocked by the default deny.

b.ii. Best situations for Scenario 2a

This fits a case where one trusted outside host, such as an administrator's machine, needs to reach several internal systems. It is still fairly broad because that one host is allowed to use any protocol and reach any inside host, so a rule that also limits the service or port would be safer.

c.i. Scenario 2b: what is permitted?

All IP traffic from any outside source is permitted, but only if the destination is the single host 192.168.200.10. Any other internal host stays protected by the default deny.

c.ii. Best situations for Scenario 2b

This is useful when one internal server is meant to be publicly reachable for many kinds of traffic, such as a general test server. For a normal production server, it is usually better to only permit the exact services needed, instead of every IP protocol.

d.i. Scenario 3a: what is permitted?

All TCP traffic from any outside source to any destination is permitted. This lets TCP based services such as SSH and HTTP through, but ICMP (ping) is not covered by this rule, so ping stays blocked.

d.ii. Scenario 3a compared with Scenario 1

Scenario 3a is narrower than Scenario 1 because it only opens TCP traffic, not all IP protocols. This blocks things like ICMP, which reduces some risk, but it is still a very wide rule because every TCP port and every destination is open.

e.i. Scenario 3b: what is permitted?

All ICMP traffic from any outside source to any destination is permitted. This allows ping to work, but SSH and HTTP stay blocked because they use TCP, not ICMP.

e.ii. Best situations for Scenario 3b

This can be useful when only basic reachability testing with ping is needed from outside. In a real production network, it is usually better to restrict which ICMP types and which source or destination addresses are allowed, instead of opening ICMP for everyone.

f.i. Scenario 4a: what is permitted?

Only TCP traffic from the specific host 172.16.200.10 to the 192.168.200.0/24 subnet is permitted, and only when the destination port is SSH (TCP 22). Ping and HTTP are not covered, so they stay blocked.

f.ii. Best situations for Scenario 4a

This fits a trusted external administrator who needs SSH access to manage several internal machines. It is close to least privilege, since the source host, destination subnet, protocol, and port are all limited.

g.i. Scenario 4b: what is permitted?

TCP traffic from any outside source to the internal host 192.168.200.10 is permitted, but only on port 80 (HTTP). Ping and SSH to that host stay blocked unless separate rules are added.

g.ii. Best situations for Scenario 4b

This matches the common case of publishing one internal web server so outside users can browse it. It is safer than allowing all traffic to that host, because only the web port is opened.

h.i. Scenario 5a: what is permitted?

Since permit ip any any is listed first, all IP traffic is permitted. The deny rule for HTTP to 192.168.200.10 that comes after it is never actually used, because the permit rule above it already matches first. So HTTP still gets through, even though the intention was to block it.

h.ii. Scenario 5a compared with scenarios 2 to 4

Scenarios 2 to 4 build the rule by only allowing exactly what is needed, and everything else is blocked by the default deny at the end of the list. Scenario 5a tries the opposite approach, allow everything, then block one thing, but places the rules in the wrong order. Because ASA checks rules from top to bottom and stops at the first match, the broad permit rule hides the deny rule that comes after it.

h.iii. Best situations for Scenario 5a

As written, it does not work for blocking HTTP, so it is not suitable here. If the deny rule was placed before the permit rule, this allow-most-but-block-a-few approach could make sense in a network where almost everything from outside is meant to be allowed except a small list of exceptions. In general, this is less safe than only allowing what is needed.

i.i. Scenario 5b: what is permitted?

The first rule denies TCP/SSH traffic from any outside source to 192.168.200.10. The second rule, permit ip any any, then allows everything else. So SSH to that host is blocked, but ping and HTTP to it, and traffic to any other host, are allowed.

i.ii. Compare Scenario 5b with Scenario 5a

Scenario 5b works as intended because the specific deny rule is placed before the broad permit rule, so SSH gets blocked before the permit rule has a chance to allow it. Scenario 5a fails because the order is reversed. This shows that the order of rules in an ASA access list matters a lot. A specific exception must be placed above a broader rule, or the broader rule will match first and the exception will never be used.

7. Conclusion

This lab showed how the Cisco ASA uses security levels and stateful inspection to filter traffic by default, and how access lists can change that behavior. The main lesson is that rules should only allow what is actually needed (source, destination, protocol, and port should all be as specific as possible), and that ASA reads access list entries from top to bottom, stopping at the first match. Because of this, specific rules must always be placed above more general rules for them to have any effect.